Quantum sampling for a long/short stock split:
a same-budget benchmark of QAOA variants on 12–29 qubits

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Abstract

We encode the choice of a low-risk long/short (or buy/skip) stock portfolio as an Ising problem with one qubit per asset, enumerate the exact optimum at 12 assets, and benchmark seven circuit families (linear-ramp QAOA, free-angle QAOA with five optimisers, CVaR, multi-angle, counterdiabatic, two imaginary-time methods and a constraint-preserving XY mixer) on an exact statevector simulator against classical heuristics *under the same budget*: 1 000 measured shots versus 1 000 evaluated candidate portfolios. Tuned circuits find the optimum in 99–100 % of runs, restarted greedy descent in 82 and 100 % (modes A and B), a 1 000-step annealing run in 47 and 100 %, random sampling in 22 %. The circuits, however, first cost 300–10 700 offline simulated circuits to tune, a price the heuristics do not pay, and at 12 qubits 1 000 greedy evaluations already cover a quarter of the search space. Two findings are robust: the linear-ramp recipe as commonly written has the wrong sign for the $R_x(+2\beta)$ mixer and steers towards the worst portfolios, and angles tuned at 12 qubits transfer to 16, 20 and 29 qubits without loss. An exact 29-qubit simulation (all 2^{29} energies enumerated, so the true optimum is known) shows the untuned circuit sampling with quality 0.96 (A) and 0.98 (B); within the same 1 000-evaluation budget its best portfolio (quality 0.995 in mode A) is slightly better than restarted greedy (0.993) and annealing (0.992), while no sampler reaches the exact optimum, one state in 537 million, reliably. We also measure the all-to-all connectivity advantage (66 vs. 285 two-qubit gates per layer at 12 qubits, 244 vs. 1 576 at 29). All optimised portfolios reduced out-of-sample risk; none beat the market's return. No quantum advantage is claimed and no hardware run is reported.

1 Problem and data

Each asset i gets a spin $s_i \in \{+1, -1\}$. Mode A (long/short): $+1$ = buy, -1 = short, weights s_i/n. Mode B (buy/skip): $x_i = (1 + s_i)/2 \in \{0, 1\}$, weight x_i/n, the rest in cash. With training-window mean weekly log returns μ and sample covariance C, the score to minimise is

$$J = \frac{w^{\mathsf{T}} C\, w}{V_{\mathrm{ref}}} - \eta\, \frac{\mu^{\mathsf{T}} w}{\sigma_{\mathrm{ref}}}, \qquad V_{\mathrm{ref}} = \overline{\mathrm{diag}\, C},\ \sigma_{\mathrm{ref}} = \sqrt{V_{\mathrm{ref}}}, \tag{1}$$

with $\eta \in \{0.25, 0.5, 1, 2\}$ ($\eta = 1$ default) and an optional balance constraint (equal numbers on each side). J is an Ising energy $\sum_{i<j} J_{ij} s_i s_j + \sum_i h_i s_i + c$ exactly (verified to 10^{-16} on random states); thinning the 29-asset couplings to the strongest 60 % by $|\rho_{ij}|$ gives a surrogate within 0.013 of the true score (rank correlation 0.975), and every candidate is re-scored on the dense objective.

Data. Kaggle "Huge Stock Market Dataset" (CC0): 29 named US assets in 10 groups (tech, gold & silver, energy, banks, bonds, healthcare, consumer staples, utilities, real estate, copper) plus SPY as a market reference, weekly closes 2006-05-26 to 2017-11-10 (599 weeks, 598 weekly log returns). Chronological split: 418 training weeks (to 2014-05-30, containing the 2008 crash) and 180 held-out weeks; nothing was chosen with test data. The 12-asset instance takes one asset per group in a fixed round-robin: AAPL, GLD, XOM, JPM, TLT, JNJ, PG, XLU, VNQ, FCX, MSFT, SLV. Bonds are the only group negatively correlated with the rest (mean $\rho \approx -0.22$; TLT–XLK -0.45); gold is near zero ($+0.10$ to $+0.14$). An earlier thinning rule by $|C_{ij}|$ was dropped because it removed every bond link.

2 Methods

Answer key and classical baselines. At 12 assets all $2^{12} = 4\,096$ portfolios are enumerated (exactly one optimum per setting). Baselines, written in NumPy: uniform random sampling, greedy single-flip (or swap) descent with restarts, and simulated annealing. Balance is enforced by a penalty $\lambda(\sum_i s_i)^2$ at 12 assets, with λ verified by enumeration so that no unbalanced state beats the best balanced one, and by a hard constraint in the solvers at 29.

Circuit. $H^{\otimes n}$, then per layer k: $R_{zz}(2\gamma_k J_{ij}/\mathrm{scale})$ on every kept link, $R_z(2\gamma_k h_i/\mathrm{scale})$ on every asset, $R_x(2\beta_k)$ on every qubit, with one common scale $= \max(|J_{ij}|, |h_i|)$; the constant is a global phase. Linear-ramp QAOA (LR-QAOA) [1] uses $\gamma_k = \frac{k}{p}\Delta\gamma$, $\beta_k = (1 - \frac{k-1}{p})\Delta\beta$ and was grid-tuned (300 points per p, $p = 1..5$) by lowest expected energy, never by the optimum's probability, because the optimum is unknown in a real problem. Free-angle QAOA ($2p$ angles) was tuned by COBYLA, Nelder–Mead, Powell, L-BFGS-B and our own SPSA from the same five starts (the default ramp, the tuned ramp, three fixed-seed random points) with a cap of $200p$ circuit evaluations each. Extras at the same cap: a CVaR objective ($\alpha = 0.1, 0.25$), multi-angle QAOA ($3p$: separate angles for links and fields), a first-order counterdiabatic R_y layer (DC-QAOA, cf. [3]), VarQITE angle training (McLachlan, central differences), varQITE with the Pauli-term update rule of IonQ/ORNL [2] (ansatz A: R_y + CX per link, 66 parameters; ansatz B: R_{zz} per link + R_x layer; implemented from the method description, no code reused), and an XY mixer with a Dicke initial state for the balance constraint. An independent

NumPy statevector simulator (1.7 ms per $p = 2$ circuit at 12 qubits) agrees with Qiskit 2.5.2 `Statevector` to 3×10^{-16} for all variants; the R_{zz}/R_z sign convention is unit-tested against the explicit diagonal unitary. 122 unit tests pass; the full benchmark ran in 121 s on 8 cores.

Same-budget protocol. One *candidate evaluation* is the energy of one portfolio: a measured shot, a random sample, one neighbour examined by greedy descent, or one annealing proposal. Every method gets exactly $B = 1\,000$ of them; hit@1000 is the probability that the optimum is found within the budget (exact $1 - (1 - P_{\text{opt}})^{1000}$ for circuits and confirmed by 200 sampled repetitions; 200 repetitions for the heuristics). Greedy restarts from random states until the budget is used (19 and 12 restarts on average in modes A and B); annealing runs one geometric schedule of 1 000 proposals. The circuits' offline tuning cost is listed separately.

3 Same-budget benchmark

Table 1 and Fig. 1 give the fair comparison. With 1 000 evaluations each, the tuned circuits reach the optimum in 99–100 % of runs in both modes; restarted greedy reaches 82 % (A) and 100 % (B), the 1 000-step annealer 47 % and 100 %, uniform sampling 22 %. Three caveats keep this honest: (i) the circuits were first tuned with 300 (LR-QAOA) to 10 685 (free-angle, all optimisers and starts) simulated circuits, which the heuristics never needed; (ii) at 12 qubits 1 000 greedy evaluations already cover a quarter of the 4 096 states, and the annealer's schedule was not tuned for such a short run; (iii) all distributions are exact and noiseless. Mode B is easy for every method because its optimum ("buy AAPL and TLT", everything else in cash) is reached by few flips from most states; mode A, a genuine hedge, separates the methods.

4 Per-shot results and the two findings

Per shot, the tuned circuits are 30× (A) and 85× (B) more likely than uniform sampling to return the optimum (Table 1); the CVaR objective reaches 236× in mode B at a lower mean quality. In mode A the circuit's single most likely portfolio *is* the optimum (rank 1 of 4 096: long AAPL, MSFT, GLD, SLV, JPM, TLT, PG, XLU; short XOM, JNJ, VNQ, FCX; Fig. 2); in mode B it is "buy TLT only", one asset from the optimum (rank 6).

Finding 1: the sign of the ramp. With the mixer $R_x(+2\beta)$ and $H^{\otimes n}$ as the initial state, lowering the energy requires γ and β of *opposite* sign. The recipe as commonly written ($\Delta\gamma = 0.6$, $\Delta\beta = 0.3$) drives the state towards the *highest* energies: quality 0.45 at $p = 5$ against 0.81 for uniform sampling in mode A, 0.33 against 0.71 in mode B, and the probability of the worst portfolio grows with p. Fig. 3 shows the grid: the good region lies entirely in the half-plane $\Delta\beta < 0$. All tuned ramps have $\Delta\beta < 0$: A (0.3,−0.3), (0.5,−0.3), (0.7,−0.4), (0.7,−0.4), (0.7,−0.4) for p = 1..5; B (1.0,−0.7), (1.4,−0.6), (1.4,−0.6), (1.4,−0.5), (1.3,−0.5).

Finding 2: angle transfer. The ramp tuned at 12 assets keeps its quality at 16 and 20 assets (Table 2), within 0.005 of a fresh grid search at 16, and still beats uniform sampling at 20 assets thinned to 60 % of the links as the 29-asset model is. This justifies running the 29-qubit circuits with the 12-asset ramp, untuned; beyond about 20 qubits offline tuning is impossible and would have to be paid in hardware shots.

Optimisers and extras. The optimiser matters less than the start: from the tuned ramp all five reach about the same quality (mean over starts and p: A 0.857–0.895, B 0.950–0.969), random starts often stall, and L-BFGS-B needs the fewest evaluations (24–98). CVaR raises P_{opt} at the cost of mean quality; multi-angle gains little; the counterdiabatic layer in mode B concentrates on near-optimal states (quality 0.985) but never on the optimum. VarQITE angle training matches standard QAOA from the same starts. varQITE with the Pauli-term update (Fig. 4) lowers the energy steadily but slowly at the published learning rates (0.05–0.4) and reaches quality 0.880 (A) / 0.871 (B) after 60 steps and 7 980 simulated circuits per run with 66 two-qubit gates, below LR-QAOA at one layer; its IonQ-native ansatz B is a stationary point at $\theta = 0$ (every parameter-shift derivative vanishes) and needs a random start. Exact imaginary-time evolution of a diagonal cost is classical Boltzmann sampling; as a reference curve it overtakes the best LR-QAOA at $\tau = 80$ (A) / 160 (B).

5 Balance constraint in the circuit

The penalty coupling 2λ is 12× (A) / 5× (B) the largest objective term, so penalised circuits mostly learn the constraint and their grids run into the small-$\Delta\gamma$ edge. The XY mixer (12 XX+YY gates per layer on a ring, Dicke state prepared generically, not gate-optimised) keeps every shot balanced and is 10× more likely than the penalty to return the best balanced portfolio (Table 3). At 29 qubits post-selection of the unconstrained circuit is the practical choice.

6 Gate counts

On all-to-all connectivity one layer costs one native two-qubit gate per link; routed on a heavy-hex grid the same layer costs 4.3× (12 qubits) and 6.5× (29 qubits) more (Table 4). The 29-qubit circuits (732 two-qubit gates at $p = 3$) were built and counted only, never simulated.

7 Out-of-sample finance

Over the 180 held-out weeks (simple returns, equal weights, no costs), the classically optimal portfolios of all 32 settings (2 modes × 4 η × balance on/off) had lower weekly volatility than SPY (0.00–0.91 % vs. 1.57 %) and than 1 000 random portfolios of the same kind; 14 of 32 earned more than a random portfolio; none beat SPY's return (+0.165 vs. +0.214 % per week

Table 1: Same-budget benchmark at 12 assets, $\eta = 1$, balance off. hit@1000: probability of finding the optimal portfolio within 1 000 candidate evaluations. P_{opt}: probability that a single shot is the optimum (uniform: $1/4096 = 0.00024$). Quality $= (E_{\max} - \langle E \rangle)/(E_{\max} - E_{\min})$ of the sampled distribution (1 = always the optimum). Tuning: offline simulated circuits used to choose the angles (all optimisers and starts at that p). Best p is given as A / B.

		Mode A: long/short			Mode B: buy or skip		
Method (best p A / B)	tuning	hit@1000	P_{opt}/shot	quality	hit@1000	P_{opt}/shot	quality
uniform random sampling	0	0.217	0.0002	0.807	0.217	0.0002	0.714
greedy descent, restarted	0	0.815	–	–	1.000	–	–
simulated annealing, one 1000-step run	0	0.465	–	–	1.000	–	–
LR-QAOA (5 / 4)	300 / 300	0.999	0.0073	0.950	1.000	0.0207	0.973
free-angle QAOA (3 / 2)	10 685 / 6 489	0.986	0.0042	0.935	1.000	0.0122	0.976
CVaR-QAOA (α=0.1) (3 / 3)	3 047 / 4 860	0.999	0.0066	0.918	1.000	0.0577	0.948
multi-angle QAOA (3 / 2)	5 730 / 3 626	0.995	0.0054	0.945	1.000	0.0141	0.986
counterdiabatic QAOA (3 / 3)	5 154 / 5 210	0.999	0.0067	0.938	$<10^{-3}$	$<10^{-4}$	0.985
VarQITE angle training (3 / 3)	3 000 / 3 000	0.986	0.0042	0.935	1.000	0.0258	0.978
varQITE Pauli-update [2]	38 792 / 46 891	$<10^{-3}$	$<10^{-4}$	0.880	$<10^{-3}$	$<10^{-4}$	0.871

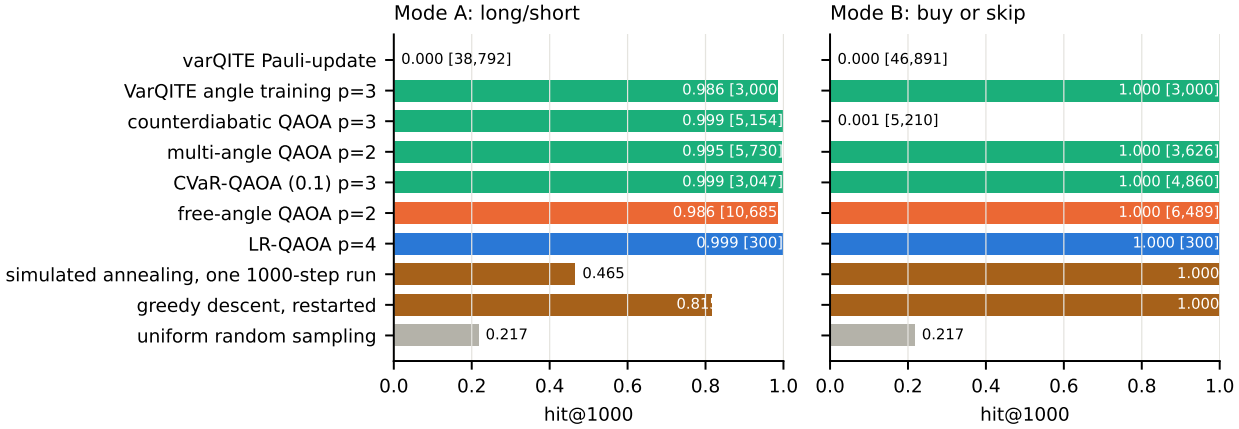


Figure 1: hit@1000 for every method under the same budget of 1 000 candidate evaluations. Brown: classical heuristics (measured over 200 repetitions); grey: uniform sampling; blue, orange, green: circuits (exact). The bracketed numbers are the offline tuning circuits.

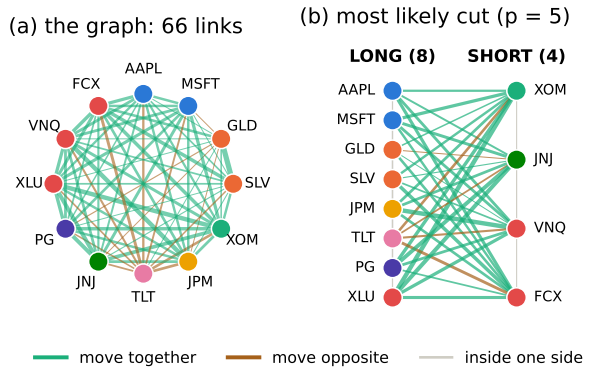


Figure 2: Mode A, 12 assets. (a) Training correlations: teal = move together, brown = move opposite, width $\propto |\rho|$. (b) The most likely LR-QAOA portfolio: 27 of 52 "together" links are cut (hedged) and only 5 of 14 "opposite" links.

in a strongly rising market). The method reduces risk; it does not raise return. Trading costs, borrow fees and slippage are not modelled, and the universe has survivorship bias.

Table 2: Transfer of the 12-asset ramp: quality of the sampled distribution at $p = 1/2/3$ with the 12-asset angles ("own": a fresh grid search at that size).

assets (links)	mode	$p = 1$	$p = 2$	$p = 3$
12 (all)	A	0.895	0.916	0.931
16 (all)	A	0.889	0.905	0.922
16 (all), own	A	0.890	0.910	0.926
20 (all)	A	0.906	0.919	0.937
20 (60 %)	A	0.864	0.882	0.900
12 (all)	B	0.955	0.969	0.972
16 (all)	B	0.949	0.960	0.963
16 (all), own	B	0.949	0.960	0.963
20 (all)	B	0.959	0.968	0.969
20 (60 %)	B	0.950	0.963	0.966

8 29 qubits: exact simulation with transferred angles

The memory-lean simulator (energies built bit by bit, complex64 amplitudes, 13 GB) makes the 29-qubit case exact: all 2^{29} energies are enumerated, which gives the true optimum and shows that the Task-3 annealer had already found it in both modes. Table 5 and Fig. 5 report LR-QAOA with the ramp transferred from 12

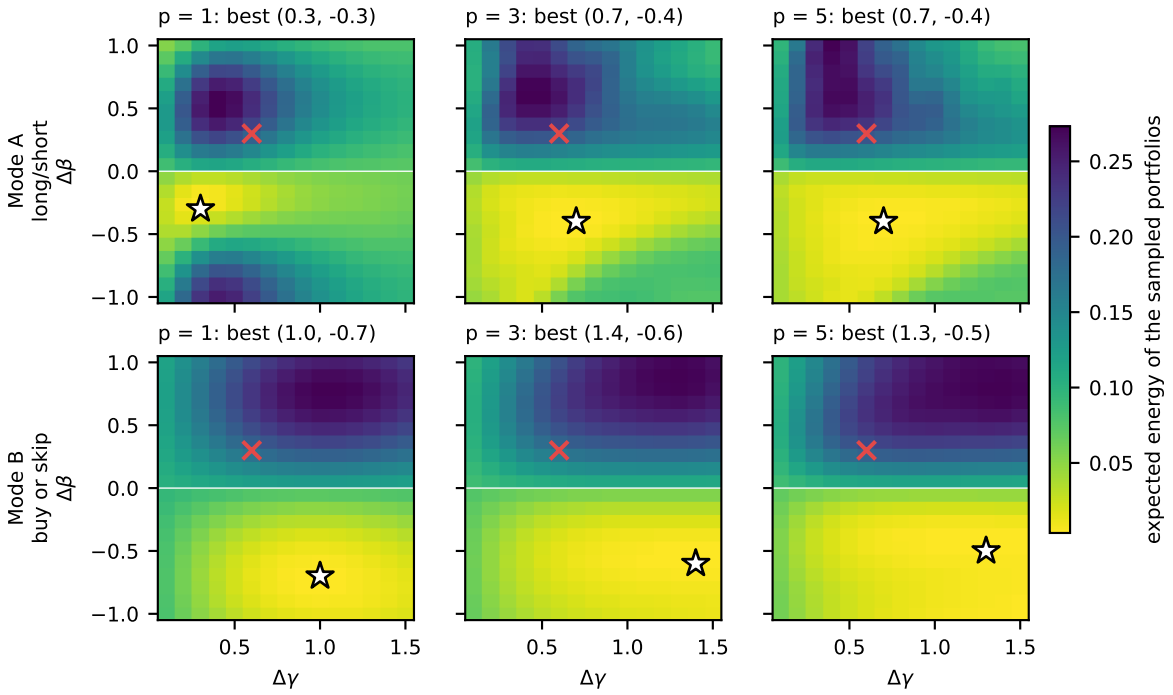


Figure 3: LR-QAOA grid search, expected energy of the sampled portfolios (darker = lower = better) over $\Delta\gamma$ and $\Delta\beta$ for $p = 1, 3, 5$. White star: chosen ramp; red cross: the recipe as commonly written (0.6, 0.3), which lies in the bright upper half where the circuit samples worse-than-random portfolios.

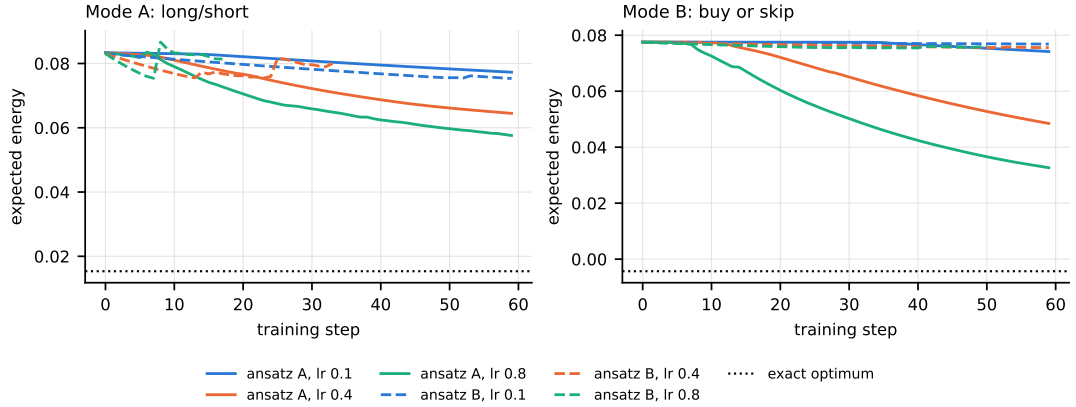


Figure 4: varQITE (Pauli-term update rule [2]) training curves on the exact statevector: expected energy per step for both ansätze (solid: A, dashed: B) and three learning rates (blue 0.1, orange 0.4, green 0.8); dotted: the exact optimum. Ansatz A learns, ansatz B barely moves from its small random start.

Table 3: Equal buys and sells at 12 assets, $p = 2$: share of balanced shots and probability per shot of the best balanced portfolio.

	Mode A		Mode B	
method	balanced	P_{best}	balanced	P_{best}
uniform, post-selected	23 %	0.0002	23 %	0.0002
penalty in the model	97 %	0.0011	97 %	0.0011
post-selection	31 %	0.0019	1 %	0.0009
XY mixer + Dicke start	100 %	**0.0192**	100 %	**0.0068**

Table 4: One QAOA layer, measured with the Qiskit transpiler (optimisation level 1, five routing seeds). All-to-all: trapped ions, where R_{zz} is native on IonQ Forte. Heavy-hex: a superconducting layout with SWAP routing (median, min–max).

qubits	links	2q gates	2q depth	heavy-hex 2q gates
12	66	66	15	285 (276–288)
29	244	244	29	1 576 (1 510–1 597)

assets. With the transferred ramp the sampled distribution reaches quality 0.958 (A, $p = 5$) and 0.979 (B, $p = 4$) against 0.876 and 0.732 for uniform sampling, and the optimum is 2271× (A) and 72065× (B) more likely per shot than under uniform sampling; at 2^{29} states that is still only 4.2×10^{-6} per shot (A), so no sampler finds the exact optimum in 1 000 shots. The quality of the best portfolio found within 1 000 evaluations is 0.9952 for LR-QAOA (A) against 0.9927 for restarted greedy descent and 0.9924 for a 1 000-step annealing run, which reach the exact optimum in 2% and 4% of runs; in mode B the figures are 0.9993 (LR-QAOA), 0.9998 (greedy, 61%) and 0.9999 (annealing, 61%). The cir-

Table 5: 29 assets, $\eta = 1$, balance off, exact statevector (complex64) on an 8-core server. The exact optimum is known from enumerating all 2^{29} energies. LR-QAOA uses the 12-asset ramp without re-tuning, on the thinned circuit model (244 links) and on the dense model (406 links); every distribution is scored on the dense objective. hit@1000 and P_{opt} refer to the exact optimum, one state in 537 million; the quality of the best portfolio found within 1 000 evaluations is reported in the text.

method	model	hit@1000	P_{opt}/shot	quality	best-of-1000	most likely = optimum
Mode A: long/short; exact optimum -0.01017; the Task-3 annealer had found it: yes						
uniform random sampling	–	0.000	1.9×10^{-9}	0.876	0.9772	–
greedy descent, restarted	dense	0.015	–	–	0.9927	–
simulated annealing, one 1000-step run	dense	0.035	–	–	0.9924	–
LR-QAOA $p = 1$ (12-asset ramp)	thinned	0.000	2.8×10^{-8}	0.919	0.9846	no
LR-QAOA $p = 2$ (12-asset ramp)	thinned	0.000	1.2×10^{-7}	0.933	0.9886	no
LR-QAOA $p = 3$ (12-asset ramp)	thinned	0.001	1.1×10^{-6}	0.947	0.9928	no
LR-QAOA $p = 4$ (12-asset ramp)	thinned	0.002	2.2×10^{-6}	0.953	0.9942	no
LR-QAOA $p = 5$ (12-asset ramp)	thinned	0.004	4.2×10^{-6}	0.958	0.9952	no
LR-QAOA $p = 1$ (12-asset ramp)	dense	0.000	1.8×10^{-8}	0.915	0.9836	no
LR-QAOA $p = 2$ (12-asset ramp)	dense	0.000	8.0×10^{-8}	0.933	0.9876	yes
LR-QAOA $p = 3$ (12-asset ramp)	dense	0.001	7.7×10^{-7}	0.946	0.9922	no
Mode B: buy or skip; exact optimum -0.00621; the Task-3 annealer had found it: yes						
uniform random sampling	–	0.000	1.9×10^{-9}	0.732	0.9652	–
greedy descent, restarted	dense	0.610	–	–	0.9998	–
simulated annealing, one 1000-step run	dense	0.610	–	–	0.9999	–
LR-QAOA $p = 1$ (12-asset ramp)	thinned	0.035	3.6×10^{-5}	0.965	0.9985	no
LR-QAOA $p = 2$ (12-asset ramp)	thinned	0.073	7.6×10^{-5}	0.975	0.9991	no
LR-QAOA $p = 3$ (12-asset ramp)	thinned	0.088	9.2×10^{-5}	0.977	0.9992	no
LR-QAOA $p = 4$ (12-asset ramp)	thinned	0.126	1.3×10^{-4}	0.979	0.9993	no
LR-QAOA $p = 5$ (12-asset ramp)	thinned	0.112	1.2×10^{-4}	0.978	0.9993	no
LR-QAOA $p = 1$ (12-asset ramp)	dense	0.036	3.6×10^{-5}	0.965	0.9986	no
LR-QAOA $p = 2$ (12-asset ramp)	dense	0.063	6.5×10^{-5}	0.973	0.9990	no
LR-QAOA $p = 3$ (12-asset ramp)	dense	0.066	6.8×10^{-5}	0.973	0.9990	no

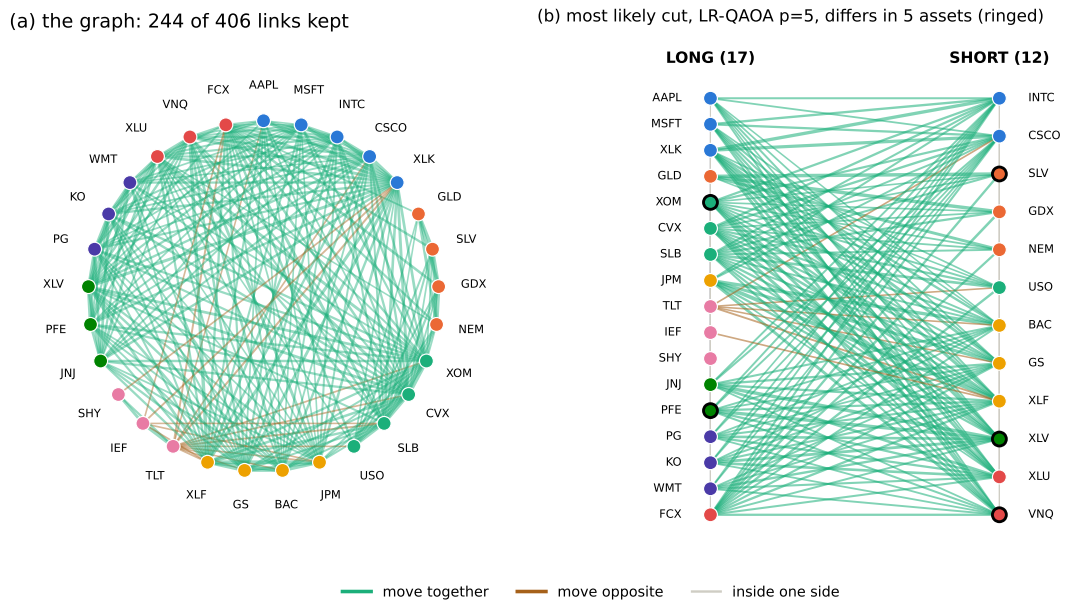


Figure 5: Mode A, 29 assets. (a) The thinned correlation graph (244 of 406 links). (b) The most likely portfolio of LR-QAOA at 29 qubits with the 12-asset ramp; assets ringed in black differ from the exact optimum.

cuit's most likely portfolio in mode A differs from the exact optimum in 5 assets (Fig. 5); the thinned surrogate's own optimum differs from the dense optimum. At this size the classical heuristics win on reaching the exact optimum within the budget; the untuned circuit still samples far above random, which is what the hardware run will test.

9 Limitations and outlook

Everything above is an exact classical simulation; the IonQ noise-model run (one batched job with the $p = 0$ control, LR-QAOA, CVaR and XY circuits; 12 qubits live, 29 recorded) is the next step and will show how much survives 66–244 two-qubit gates per layer. The same-budget comparison favours the circuits only after their offline tuning, which is impossible beyond about 20 qubits; the angle transfer is the way around it and is now measured exactly up to 29 qubits, where the circuit keeps its sampling quality but the classical heuristics

reach the exact optimum more often. Deferred: warm starts, variational real-time evolution, recursive QAOA. Code, 122 tests, the saved angles and all run records (including every unhelpful variant) are in the team repository.

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